

Agreement-Margin Abstention for Selective Question Answering: A Diagnostic Result

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Abstract

Selective question answering requires confidence scores that reject likely wrong generations while preserving useful coverage, yet generated short answers make confidence difficult because semantic agreement, token probabilities, and dataset aliases can fail in different ways. Agreement-Margin Abstention (AMA) tests a simple post-hoc fusion: it clusters sampled short answers into semantic modes, combines the largest-cluster agreement with a greedy top-token margin, and calibrates a scalar confidence score on development data without adapting model weights. We evaluate AMA on 720 held-out questions from Natural Questions Open, TriviaQA, and WebQuestions using one frozen Qwen2.5-7B/vLLM generation cache and five confidence baselines: agreement, semantic entropy, sequence logprob, predictive entropy, and margin-only confidence. AMA attains the best aggregate AURAC in this matrix, 0.563673 versus 0.558692 for semantic entropy, but the +0.004981 delta falls below the preregistered +0.01 threshold, its paired bootstrap interval crosses zero, and sensitivity analyses over sample count and clustering threshold do not make the effect stable. The supported contribution is therefore a diagnostic account of selective-QA confidence: agreement-style signals dominate margin-only confidence, but high agreement can still certify repeated wrong answers, and small fusion gains can disappear across benchmark families and clustering choices.

1 Introduction

Open-domain question answering systems increasingly need a reject option: a model should answer when it is likely correct and abstain when its generation is unreliable. This objective fits the selective prediction tradition, where systems are evaluated by the risk–coverage trade-off rather than by full-coverage accuracy alone (El-Yaniv and Wiener,

2010; Geifman and El-Yaniv, 2019). For large language models, abstention also intersects with hallucination mitigation, truthfulness measurement, and safety-oriented refusal behavior (Wen et al., 2025; Ji et al., 2023; Lin et al., 2022b; Manakul et al., 2023). Modern open-domain QA benchmarks add another complication: Natural Questions Open, TriviaQA, and WebQuestions differ in alias coverage, answer granularity, and whether a short entity or date should be credited (Joshi et al., 2017; Kwiatkowski et al., 2019; Berant et al., 2013).

Confidence estimation for generated short answers therefore cannot be reduced to one familiar scalar. Token probabilities can be sharp for fluent but wrong continuations, while sampling-based agreement can be high when the model repeatedly emits the same misconception. Recent semantic-uncertainty work motivates clustering generations before scoring confidence, but it leaves open whether a token-level signal can help when clusters are stable but wrong (Kuhn et al., 2023; Farquhar et al., 2024). The selective-QA confidence score is thus a ranking rule over generated strings, aliases, and sampling behavior, and its errors can come from model knowledge, decoding instability, or benchmark normalization.

Agreement-Margin Abstention (AMA) evaluates this gap with a deliberately simple post-hoc protocol. For each question, AMA samples short answers, clusters them into semantic modes, computes largest-cluster agreement, and combines that agreement with the average greedy top-token margin. The intended complementarity is specific: if samples cluster tightly but the greedy trace has low local margin, the model may be semantically consistent but token-unstable; if the margin is high but sampled answers disagree, a confident local continuation may not imply global answer reliability. The method reuses inference artifacts exposed by a single served model, rather than adding retrieval, an external judge, or learned model adaptation.

The evidence supports a diagnostic conclusion rather than a broad improvement claim. In a frozen held-out matrix over three public QA families, AMA is the top aggregate method by AURAC, but only by +0.004981 over semantic entropy, with a bootstrap confidence interval that includes zero. The same run shows that the lead is below the pre-registered +0.01 threshold, that family-level winners differ, and that plausible clustering changes can make agreement or semantic entropy stronger than AMA. Those facts make the result useful as a boundary case for confidence fusion: agreement is a strong signal, raw margin is weak, and combining them naively does not automatically produce a reliable abstention policy. Our contributions are:

1. a reproducible AMA protocol that fuses semantic agreement and token-margin confidence without training model weights;
2. a three-family selective-QA evaluation against agreement, semantic entropy, sequence log-prob, predictive entropy, and margin-only baselines;
3. a diagnostic analysis showing that margin-only confidence is weak, while high agreement can still produce wrong accepted answers and sensitivity settings can reverse the apparent winner.

2 Related Work

Selective prediction and calibration. Selective prediction formalizes the choice to abstain from uncertain predictions and evaluate risk as coverage changes (El-Yaniv and Wiener, 2010). Early neural selective-classification work studies reject options for deep models (Geifman and El-Yaniv, 2017), and SelectiveNet demonstrates that learned reject heads can directly optimize this trade-off in supervised settings (Geifman and El-Yaniv, 2019). Post-hoc confidence work shows both the appeal and fragility of scalar confidence transformations, maximum-softmax baselines, trust scores, and uncertainty under shift (Niculescu-Mizil and Caruana, 2005; Guo et al., 2017; Hendrycks and Gimpel, 2017; Jiang et al., 2018; Ovadia et al., 2019). AMA remains post-hoc: it does not retrain the evaluated model, and it uses development data only to z-score the margin and choose a scalar fusion weight. This distinction matters for interpreting the result. The experiment evaluates whether a cheap ranking signal can improve selective acceptance

of already generated answers, not whether a supervised rejector can be trained with task labels or whether a model can learn to express calibrated uncertainty during pretraining. Deep ensembles and calibrated structured prediction methods often improve predictive uncertainty when multiple models, structured outputs, or training perturbations are available (Kuleshov and Liang, 2015; Lakshminarayanan et al., 2017); AMA instead asks how far a single served model can go using only samples and greedy logprob traces.

Uncertainty for generated language. Sampling-based reasoning work popularized the idea that multiple generations can expose stable answer modes (Wei et al., 2022; Kojima et al., 2022; Wang et al., 2022). Semantic entropy extends this idea by clustering equivalent generations before measuring uncertainty (Kuhn et al., 2023; Farquhar et al., 2024). SelfCheckGPT similarly uses consistency across samples as evidence for hallucination detection (Manakul et al., 2023). Recent abstention work and selective-prediction studies caution that entropy-style scores alone may be insufficient for safe LLM decisions (Wen et al., 2025; Phillips et al., 2026). AMA tests whether adding a token-margin signal to semantic agreement improves this setting; the answer here is mostly no, but the failure is informative. The failure also relates to work showing that language models can sometimes report their own knowledge state or verbalize uncertainty while still producing overconfident or misleading surface forms (Kadavath et al., 2022; Lin et al., 2022a). Self-consistency, semantic entropy, and self-checking methods all depend on the diversity and fidelity of sampled outputs. If the model’s samples collapse around a wrong entity, agreement can become a false certificate; if samples are diverse because aliases or formatting vary, entropy can penalize an answer set that is semantically adequate. AMA was designed to probe whether token-level margin adds an orthogonal warning signal in those two cases. Unlike self-refinement or judge-based evaluation pipelines, AMA does not ask the same or another model to critique the answer; it only reorders a fixed generation cache (Madaan et al., 2023; Zheng et al., 2023).

Benchmarks for answerability and QA confidence. The evaluation uses three public short-answer QA families: TriviaQA (Joshi et al., 2017), Natural Questions (Kwiatkowski et al., 2019), and WebQuestions (Berant et al., 2013). These datasets

differ in answer style and alias richness, which matters for agreement clustering and normalized exact-match scoring. They sit within a broader open-domain QA literature that ranges from reading Wikipedia with retrieval to dense retrieval, retrieval-augmented generation, and closed-book parametric QA (Chen et al., 2017; Karpukhin et al., 2020; Lewis et al., 2020; Roberts et al., 2020). SQuAD-style answerability work (Rajpurkar et al., 2016, 2018) motivates abstention, but our experiment focuses on selective prediction over generated answers rather than unanswerable-question detection. Ambiguous open-domain questions further show that multiple answers or granularities can be legitimate, which is relevant to the alias-sensitive failure cases below (Min et al., 2020). Natural Questions Open contains naturally phrased information needs whose aliases are often sparse; TriviaQA has richer distant-supervision aliases and higher answerability under the short-answer prompt; WebQuestions is entity-centric and exposes date/name formatting issues. Using all three families prevents the conclusion from being driven by one alias regime. It also makes the paper’s main claim harder: an aggregate gain is not enough if the same score fails to dominate the best baseline within each family.

3 Method

Figure 1 summarizes the paper-facing AMA pipeline. For each question q , the model samples $K = 8$ short answers under the same prompt and decoding budget. Let the extracted, normalized samples be $y_1, \dots, y_K$. We cluster samples by exact normalized match and fixed embedding similarity, producing semantic clusters $C_1, \dots, C_J$. The normalization removes answer prefixes, lowercases text, and strips articles and punctuation before comparison with aliases or other generated samples. The embedding step follows the sentence-embedding tradition for semantic similarity (Reimers and Gurevych, 2019); it is intentionally frozen rather than tuned per benchmark family, with a default similarity threshold of 0.82 and reported threshold sensitivity rather than hidden tuning. This makes the cluster score reproducible but imperfect, since it can both merge distinct answers and split paraphrases. The agreement score is the relative size of the largest cluster:

$$A(q) = \max_j |C_j|/K. \quad (1)$$

AMA then obtains a greedy answer trace for the

same question. For each generated token position t , the inference backend returns the chosen-token logprob and the best alternative logprob. The token-margin score averages this local separation:

$$M(q) = \frac{1}{T} \sum_{t=1}^T \left(\log p_t^{(1)} - \log p_t^{(2)} \right). \quad (2)$$

Margin values are z-scored using only the calibration split. The final confidence score is a scalar fusion:

$$\text{AMA}(q) = \alpha A(q) + (1 - \alpha) z(M(q)). \quad (3)$$

The development split chooses α on a grid to maximize AURAC; all parameters are frozen before held-out evaluation. The learned object is therefore only a one-dimensional calibration weight, not an adapted language model. This choice keeps the diagnostic question clean: if the margin signal is useful, it should help even under a restricted post-hoc fusion; if it does not help here, stronger claims about margin-based abstention need additional evidence. The final selected setting gives most weight to agreement, which already foreshadows the result that margin contributes little stable information.

The baselines use the same cached generations and greedy traces. Agreement uses $A(q)$ alone. Margin uses $z(M(q))$ alone. Semantic entropy scores the entropy of semantic clusters (Kuhn et al., 2023). Sequence logprob uses the length-normalized greedy answer probability. Predictive entropy uses the mean token predictive entropy from the greedy trace. Because every method consumes the same outputs, comparisons isolate the confidence score rather than generation quality. All methods therefore share the same full-coverage exact-match accuracy on a given benchmark family. Differences in AURAC and fixed-coverage accuracy arise only because each confidence score orders accepted examples differently. This shared-cache design is important for paired bootstrap testing: each method sees the same questions, gold aliases, sampled answers, greedy trace, and extraction artifacts.

4 Experimental Setup

Benchmarks and model. The held-out matrix contains 720 questions: 240 from Natural Questions Open validation, 240 from TriviaQA validation, and 240 from WebQuestions test. Calibration uses non-overlapping public training splits

Agreement-Margin Abstention

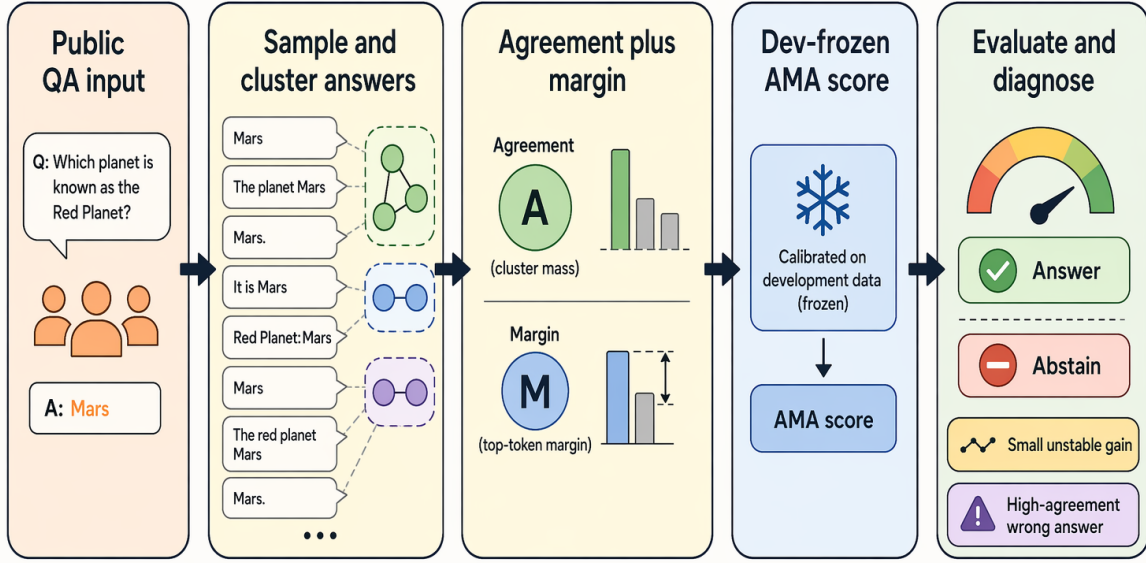


Figure 1: AMA combines sampled semantic agreement with token-margin confidence, but the held-out gain is weak: 0.563673 AURAC versus 0.558692 for semantic entropy, with a confidence interval crossing zero.

with the same per-family count. All evaluated answers come from Qwen/Qwen2.5-7B served through vLLM, whose PagedAttention-style serving design supports efficient batched generation and logprob extraction (Kwon et al., 2023). The answer prompt asks for the shortest correct answer. Sampling uses $K = 8$, temperature 0.7, top- p 0.95, and a short-answer generation cap of 16 tokens; greedy traces request top-logprob information for margin, sequence-logprob, and predictive-entropy scores.

Correctness and metrics. Answers are stripped of prompt echoes and answer prefixes, normalized by lowercasing and punctuation/article removal, and scored by exact match against all provided gold aliases. The primary metric is AURAC, the area under the rejection-accuracy curve, where higher values mean that the confidence ranking places correct answers earlier as coverage decreases. Secondary summaries report accuracy at fixed coverages and paired bootstrap confidence intervals for AMA against each baseline. The preregistered positive threshold required AMA to beat the strongest non-AMA comparator by at least +0.01 aggregate AURAC and not lose on more than one benchmark

Table 1: The held-out protocol uses 720 questions, six confidence scores, and one frozen Qwen2.5-7B/vLLM generation cache.

Component	Frozen paper-facing setting
Benchmarks	240 held-out questions each from Natural Questions Open, TriviaQA, and WebQuestions; matching non-overlapping calibration splits.
Backend	Qwen/Qwen2.5-7B with vLLM generation; same sampled and greedy traces reused by every method.
Decoding budget	$K = 8$ samples, temperature 0.7, top- p 0.95, 16-token cap, and greedy top-logprob traces.
Comparators	agreement, semantic entropy, sequence logprob, predictive entropy, margin-only confidence, and AMA.
Decision rule	Development split chooses α and freezes all scoring parameters before the 720-question held-out matrix.

family. We use this threshold as a claim gate rather than as a tuning objective. The threshold protects against treating tiny ranking changes as a substantive advance when the curves are visually close and the same model answers are being reordered. Bootstrap intervals are paired by question so that

the uncertainty reflects confidence-score ordering differences rather than independent resampling of unrelated examples. The analysis also reports sensitivity to K and clustering threshold because the main mechanism depends directly on sampled answer clustering.

Execution and artifact checks. The full run evaluates 720 held-out questions and produces 4,320 method rows, one row for each method-question pair. The calibration cache contains 720 feature rows, and the evaluation cache contains 720 feature rows. All evaluation rows include the logprob traces needed by sequence logprob, predictive entropy, and margin-based scoring. Correctness is nonconstant within each benchmark family, which rules out a degenerate scorer or a task-loading failure. The run uses a frozen calibration outcome with $\alpha = 0.95$, $\tau = 0.3288876457246317$, and development AURAC 0.524311 before held-out scoring. These implementation checks are not paper claims by themselves, but they matter because an inconclusive effect could otherwise be explained by a broken confidence path.

Stopping rule and scope. The experiment follows a smoke-pilot-full progression, but only the full held-out matrix is used for paper-facing results. Smoke rows verified that vLLM returned sampled answers and top-logprob traces; pilot rows were used only for runtime sanity checks. No synthetic questions, duplicated prompts, relabeled examples, or generated gold answers enter the final tables. The paper also does not train or adapt model weights. This keeps the contribution narrowly scoped to post-hoc selective confidence under a fixed open-weight QA backend. The downside of that scope is equally important: the evidence cannot support claims about retrieval-augmented QA, long-form generation, or learned abstention policies.

5 Main Results

Table 2 is the central result matrix: it exposes the benchmark family, split size, evaluated backend, decoding budget, metric, and all method scores in one place. AMA reaches 0.563673 aggregate AURAC, while semantic entropy reaches 0.558692. This is the best aggregate score in the run, but it is not a paper-positive effect under the preregistered criterion. The family rows further weaken a broad improvement claim: AMA trails sequence logprob

on Natural Questions Open and WebQuestions, and trails semantic entropy on TriviaQA, even though it can improve some aggregate coverage points. The stable result is instead that agreement-style scores are much stronger than token-margin alone. Because all methods share the same generated answers, the full-coverage accuracy is 0.358333 in aggregate. The table therefore measures selective ordering, not answer generation. At 80% coverage AMA accepts a set with 0.435764 accuracy, slightly above agreement at 0.432292 and semantic entropy at 0.430556. At 50% coverage AMA reaches 0.572222, again only marginally above semantic entropy at 0.569444. These fixed-coverage differences are directionally consistent with the aggregate AURAC lead, but they are too small to justify stronger language.

The family breakdown is more damaging to the original hypothesis than the aggregate row alone. On Natural Questions Open, sequence logprob reaches 0.417601 AURAC while AMA reaches 0.398383. On TriviaQA, semantic entropy reaches 0.808303 while AMA reaches 0.795103. On WebQuestions, sequence logprob reaches 0.393194 while AMA reaches 0.389234. This pattern suggests that the useful confidence signal changes with benchmark style: alias-rich TriviaQA benefits from semantic clustering, while the lower-accuracy Natural Questions and WebQuestions slices can make local sequence probability competitive.

Paired bootstrap tests confirm the interpretation; Appendix Table 4 gives the full interval table. AMA reliably beats the weakest probability-derived baselines, but not the strongest agreement-style alternatives. The conclusion is not that fusion is useless; rather, the available margin feature does not add enough stable ranking information beyond semantic agreement and entropy. The paired tests sharpen this distinction. AMA’s aggregate delta over margin-only confidence is +0.126737 AURAC with a 95% interval from +0.095439 to +0.159033, and its delta over predictive entropy is +0.071208 with an interval from +0.044782 to +0.097172. Those comparisons show that agreement-derived information is genuinely valuable. By contrast, the intervals against agreement, semantic entropy, and sequence logprob cross zero. The result is therefore not a generic failure of selective scoring; it is a failure of this particular agreement-plus-margin fusion to beat the strongest cheap alternatives robustly.

Table 2: Across three 240-question QA families and six confidence scores, AMA is only the aggregate AURAC winner; every individual family has a non-AMA winner under the same Qwen2.5-7B/vLLM, $K = 8$, 16-token budget.

Benchmark source family	/ Split / n	Model / back-end	Budget / decoding	Agree.	Sem. entr.	Seq. logp.	Pred. entr.	Margin only	AMA
All families	Held-out / 720	Qwen2.5-7B / vLLM	AURAC; $K = 8$, $T = .7$, $p = .95$, 16 tok.	0.558326	0.558692	0.547938	0.492466	0.436936	0.563673
Natural Questions Open	Val. 240	/ Qwen2.5-7B / vLLM	Same	0.377492	0.377125	0.417601	0.390442	0.334330	0.398383
TriviaQA	Val. 240	/ Qwen2.5-7B / vLLM	Same	0.806798	0.808303	0.798679	0.703164	0.614475	0.795103
WebQuestions	Test 240	/ Qwen2.5-7B / vLLM	Same	0.387262	0.387870	0.393194	0.360565	0.309647	0.389234

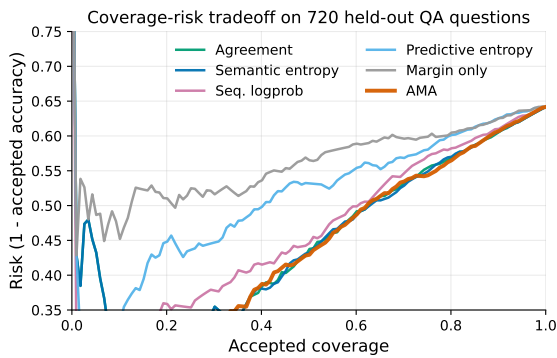


Figure 2: Coverage-risk curves show AMA tracks agreement-style baselines closely; the headline delta is too small to support a robust positive claim.

6 Analysis and Ablations

Coverage-risk behavior. Figure 2 plots the risk-coverage curves from the held-out matrix. The figure shows why aggregate AURAC differences are small: the top agreement-style curves stay close across most coverage levels, and improvements at one coverage point can disappear at another. The curves are also a reminder that a selective-QA paper should not report only one coverage threshold. If an application wants high coverage, all methods accept many uncertain answers and the accuracy gaps are compressed. If an application wants low risk at low coverage, the top of the confidence ranking dominates, and a few high-agreement wrong answers can hurt disproportionately. AMA’s aggregate advantage comes from small local reorderings, not from a visibly separated frontier.

Sample-count sensitivity. Using prefixes of the cached $K = 8$ samples, AMA has AURAC 0.528714 at $K = 2$, 0.550181 at $K = 4$, and

0.563673 at $K = 8$. At $K = 2$, sequence logprob is the strongest baseline at 0.547938, so AMA loses by -0.019224. At $K = 4$, AMA’s lead is only +0.002244. Thus the fusion needs more samples to become competitive, and even then the effect remains below the preregistered threshold.

Clustering-threshold sensitivity. The default clustering threshold is 0.82. At threshold 0.79, AMA nearly reaches the preregistered margin with a +0.009992 delta against agreement. At 0.85 and 0.88, however, agreement or semantic entropy overtake AMA. The threshold-0.88 setting gives agreement 0.590354 AURAC and AMA 0.585418. This reversal suggests that better semantic clustering can strengthen the baselines as much as the proposed fusion.

Agreement-margin regimes. Regime diagnostics split examples by family-specific agreement and margin medians. Across all families, high-agreement/high-margin examples have 0.549 accuracy, high-agreement/low-margin examples have 0.453 accuracy, low-agreement/high-margin examples have 0.159 accuracy, and low-agreement/low-margin examples have 0.112 accuracy. The separation is meaningful: low agreement is a stronger warning sign than low margin, but the high-agreement bins are still far from perfectly reliable.

What the ablations say about the original thesis. The original positive thesis required three things to be true at once: agreement should be strong, margin should add complementary signal, and the combination should be stable under reasonable analysis choices. The evidence supports only the first part: agreement and semantic entropy are consistently competitive, margin-only confidence

is much worse than agreement-style confidence, and margin separates some regimes without improving the ranking reliably. For this reason, the paper frames AMA as a diagnostic protocol rather than as a broadly competitive abstention method.

Why the margin signal is not enough. The margin feature is local: it measures how strongly the model prefers each chosen token over the next alternative along the greedy answer trace. For short answers, that trace may become confident after the model has already selected the wrong entity class or date. This helps explain why margin-only confidence has aggregate AURAC 0.436936, far below agreement and semantic entropy. It also explains why high margin cannot rescue low agreement: when samples disagree semantically, a confident greedy path is only one mode among several plausible generations.

7 Failure Cases

High-agreement failures reveal why the main result is inconclusive rather than simply positive. Several examples have agreement 1.0 and semantic entropy 0.0 but are still scored wrong under the dataset aliases. For example, a WebQuestions item asks when Truman became president; all samples answer “April 12, 1945,” while the gold alias is formatted as “4/12/1945.” This is likely an alias-normalization artifact. Other cases are more substantive: a Natural Questions Open item about Pakistan’s first elected president repeatedly answers Muhammad Ali Jinnah, while the gold aliases include Iskander Mirza. These examples show two distinct failure modes. Some high-agreement errors are evaluation-alias mismatches; others are true repeated misconceptions. AMA cannot distinguish them with agreement and token margin alone. The extracted failure table contains additional variants of both types. For WebQuestions, examples such as “1906” versus “1906 San Francisco earthquake” and “Sonny Bono” versus an alias list containing “Sonny” show that exact-match credit can depend on granularity. For TriviaQA, “Lake Placid, New York” is marked against aliases centered on “Lake Placid,” while the generated answer is arguably more specific than the canonical alias. Those cases should not be read as pure model hallucinations. They instead show that selective confidence scores are entangled with the answer-normalization contract.

Other rows are harder to excuse as alias artifacts.

Table 3: Three high-agreement failure modes explain why agreement is useful but insufficient.

Failure type	Diagnostic takeaway
Alias or granularity mismatch	Exact match can under-credit plausible high-agreement answers.
Stable misconception	Agreement and low semantic entropy can certify a repeated model error.
Mixed sample variants	Clustering choices can shift both AMA and agreement-style baselines.

The question about the theme song for *The Man with the Golden Gun* repeatedly produces Shirley Bassey, while the gold aliases identify Lulu. The Pakistan president example similarly shows a stable but wrong historical entity. These repeated misconceptions are exactly where agreement-style confidence is dangerous: the model is internally consistent, the semantic entropy is low, and the margin can be high, but the accepted answer is still wrong. An abstention mechanism that only observes the model’s own samples has limited evidence for rejecting such cases.

The diagnostic lesson is not that exact match is unusable or that agreement is unhelpful. Rather, high agreement should be interpreted as a necessary but insufficient condition for safe acceptance. When benchmark aliases are incomplete, agreement may be unfairly penalized; when the model’s misconception is stable, agreement may be overconfident. Separating those two situations likely requires evidence outside the current AMA feature set, such as retrieval-backed verification (Chen et al., 2017; Karpukhin et al., 2020; Lewis et al., 2020), entailment-based alias expansion, or supervised calibration on richer correctness labels (Min et al., 2020). Those extensions are outside this completed matrix, so they remain future work rather than implied claims.

The failure cases also clarify why a diagnostic negative result is more useful than a leaderboard-style presentation of the small aggregate gain. If one reports only the aggregate AURAC row, the reader might infer that the fused score has learned a generally better notion of answer reliability. The examples show the opposite. AMA inherits the strengths and blind spots of its two inputs: it rewards semantic concentration, and it gives a small preference to locally decisive greedy traces. Neither input checks whether the concentrated answer

is externally true. Consequently, the method can rank repeated misconceptions highly, especially when the benchmark gold set cannot expose the distinction between near-miss aliases and genuinely wrong answers. This limitation is exactly why the abstract and conclusion avoid the stronger claim that AMA improves selective QA. Appendix Table 5 preserves detailed representative patterns from the held-out cache without forcing sentence-length examples into the main-body table.

8 Conclusion

AMA provides a compact diagnostic for selective QA, but this completed matrix does not justify a robust improvement claim. The strongest supported result is narrower: semantic agreement and entropy remain hard baselines, margin-only confidence is weak, and small fusion gains can vanish under bootstrap, sample-count, and clustering-threshold scrutiny.

Limitations

The experiment uses one 7B base model and three short-answer QA families. It does not train model weights, add retrieval, or evaluate long-form generation. The exact-match scorer can under-credit aliases, while embedding-based clustering can merge or split semantic variants incorrectly. Sensitivity analysis covers cached K prefixes and clustering thresholds, not a full search over larger sample counts or alternative entailment-based clustering. The matrix also fixes one prompt and one decoding budget, so the result should be read as a controlled diagnostic for this served short-answer setting rather than as evidence about instruction-tuned systems, retrieval-augmented QA, or domains where answer correctness requires paragraph-level support. Because AMA is post-hoc and does not inspect external evidence, it cannot tell whether high agreement reflects true knowledge, alias-list incompleteness, or a repeated misconception.

Ethical Considerations

Selective QA is safety-relevant because abstention can reduce unsupported answers, but overstating small confidence gains would create a false sense of reliability. For that reason, this draft frames the primary AMA effect as weak and inconclusive. The evaluated benchmarks are public QA datasets; no private user data or human-subject collection is introduced by this work. The main practical

risk is false acceptance: a system can look safer because it rejects some low-confidence answers while still accepting repeated high-confidence errors. The held-out failures make that risk concrete, especially for short-answer settings where aliases, dates, and entity granularity affect exact-match credit. Any deployment-oriented use of these signals would need external verification, stronger calibration evidence, and domain-specific auditing before treating abstention as a safety mechanism.

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A Reproducibility Notes	792
The paper-facing reproduction target is the held-out selective-QA matrix described in the setup: three public QA families, 240 held-out questions per family, Qwen/Qwen2.5-7B generation through vLLM, six confidence methods, and AURAC plus paired bootstrap summaries. The supplementary package stores raw scored rows, derived result tables, and figure metadata so that every number in the body can be traced to its source data. No private API endpoints, credential paths, local cache paths, or device identifiers are part of the paper-facing method.	793
B Additional Tables	794
Table 4 reports the paired-bootstrap comparisons summarized in the body.	795
Table 5 gives representative high-agreement wrong rows from the held-out cache.	796
The supplementary materials also include per-family accuracy at fixed coverages, the K-prefix sensitivity rows, clustering-threshold sensitivity	797

Table 4: Paired bootstrap tests over 720 held-out questions show AMA is not significantly better than the strongest agreement-style baselines.

Comparison	AMA	Baseline	Delta	95% CI	<i>p</i>	Interpretation
Agreement	0.563673	0.558326	+0.005347	[-0.013214, 0.035041]	0.388	not significant or mixed
Semantic entropy	0.563673	0.558692	+0.004981	[-0.012145, 0.037004]	0.410	not significant or mixed
Sequence logprob	0.563673	0.547938	+0.015735	[-0.007656, 0.036905]	0.200	not significant or mixed
Predictive entropy	0.563673	0.492466	+0.071208	[0.044782, 0.097172]	0.000	reliable positive
Margin only	0.563673	0.436936	+0.126737	[0.095439, 0.159033]	0.000	reliable positive

Table 5: Representative high-agreement wrong answers show both alias/granularity artifacts and stable misconceptions.

Family	Question pattern	Majority answer	Gold alias pattern	Agreement
WebQuestions	Truman presidency date	April 12, 1945	4/12/1945	1.00
Natural Questions	Pakistan’s first elected president	Muhammad Ali Jinnah	Iskander Mirza	1.00
Open TriviaQA	Theme song for <i>The Man with the Golden Gun</i>	Shirley Bassey	Lulu	1.00
TriviaQA	Winter Olympics held in 1932 and 1980	Lake Placid, New York	Lake Placid	1.00

rows, and high-agreement failure examples. These files are sufficient to reproduce the numerical comparisons and diagnostic examples reported in the paper.